



Advanced Traffic (20–551)
Class Project Part 1

Part 1: Data Handling & Network Basics

Goals:

1. Basic Python.
2. Object-Oriented Programming (OOP) [work with classes in Python].
3. Libraries: pandas, Matplotlib, and NetworkX.
4. Read data files.
5. Visualize networks.

Tasks

1. Read the network file to define the network [Note: length dimension is meter].
2. Read node (x, y) positions to plot the network accurately.
3. Assign attributes to nodes and edges for future use.

Output

1. One or more Python files that build the network with all attributes and data (including travel time) and render it correctly. [Note: speed=80km/h]
2. A brief report on the code (inline documentation plus a README.md).
3. A short presentation of the code with Q&A.

Time

1.5 Week (1404/08/18)

Grade Weight

10% (+2.5% for the **best** set of graph attributes)